

1. Relational Algebra (Employee & Department)

Selection (σ)

σ Salary > 50000 (Employee)

Projection (π)

π Name, Salary (Employee)

Union ($\cup$)

π DeptID (Employee) $\cup$ π DeptID (Department)

Set Difference ($-$)

π DeptID (Department) - π DeptID (Employee)

Cartesian Product ($\times$)

Employee $\times$ Department

Natural Join ($\bowtie$)

Employee $\bowtie$ Department

2. CREATE TABLE Department (DeptID INT PRIMARY KEY, DeptName VARCHAR(50) NOT NULL UNIQUE);

CREATE TABLE Student (SID INT PRIMARY KEY, Name VARCHAR(50) NOT NULL, DOB DATE, DeptID INT, FOREIGN KEY (DeptID) REFERENCES Department(DeptID));

```
mysql> select * from Student;
+----+-----+-----+-----+
| SID | Name  | DOB       | DeptID |
+----+-----+-----+-----+
| 1   | Riyaz | 2005-02-10 | 101    |
| 2   | Arjun | 2004-08-15 | 102    |
| 3   | Meena | 2005-01-20 | 101    |
| 4   | Sara  | 2004-12-05 | 103    |
+----+-----+-----+-----+
4 rows in set (0.01 sec)
```

3.

```
mysql> SELECT DeptID, COUNT(*) AS TotalSales,  
-> SUM(Amount), AVG(Amount), MAX(Amount), MIN(Amount)  
-> FROM Sales  
-> GROUP BY DeptID  
-> HAVING SUM(Amount) > 5000;
```

DeptID	TotalSales	SUM(Amount)	AVG(Amount)	MAX(Amount)	MIN(Amount)
101	2	12000	6000.0000	7000	5000
102	2	7000	3500.0000	4000	3000
103	1	9000	9000.0000	9000	9000

3 rows in set (0.02 sec)

4.

```
mysql> SELECT Name  
-> FROM Employee e  
-> WHERE Salary > (  
->     SELECT AVG(Salary)  
->     FROM Employee  
->     WHERE DeptID = e.DeptID  
-> );
```

Name
Priya
Kiran

2 rows in set (0.01 sec)

5.

```
mysql> SELECT e.Name, p.ProjectName
-> FROM Employee e INNER JOIN Project p
-> ON e.EmpID = p.EmpID;
```

Name	ProjectName
John	AI System
David	Web App
Priya	ML Model
Ali	IoT System

4 rows in set (0.01 sec)

```
mysql> SELECT e.Name, p.ProjectName
-> FROM Employee e LEFT JOIN Project p
-> ON e.EmpID = p.EmpID;
```

Name	ProjectName
John	AI System
David	Web App
Priya	ML Model
Ali	IoT System
Kiran	NULL

5 rows in set (0.01 sec)

```
mysql> SELECT e.Name, p.ProjectName
-> FROM Employee e RIGHT JOIN Project p
-> ON e.EmpID = p.EmpID;
```

Name	ProjectName
John	AI System
David	Web App
Priya	ML Model
Ali	IoT System

4 rows in set (0.00 sec)

```
mysql> SELECT e.Name, p.ProjectName
-> FROM Employee e CROSS JOIN Project p;
```

Name	ProjectName
John	IoT System
John	ML Model
John	Web App
John	AI System
David	IoT System
David	ML Model
David	Web App
David	AI System
Priya	IoT System
Priya	ML Model
Priya	Web App
Priya	AI System
Ali	IoT System
Ali	ML Model
Ali	Web App
Ali	AI System
Kiran	IoT System
Kiran	ML Model
Kiran	Web App
Kiran	AI System

20 rows in set (0.01 sec)

6.

```
mysql> CREATE VIEW DeptSalaryView AS
-> SELECT d.DeptName,
-> COUNT(e.EmpID) AS TotalEmployees,
-> AVG(e.Salary) AS AvgSalary
-> FROM Department d
-> LEFT JOIN Employee e ON d.DeptID = e.DeptID
-> GROUP BY d.DeptName;
Query OK, 0 rows affected (0.04 sec)
```

```
mysql> SELECT * FROM DeptSalaryView;
+-----+-----+-----+
| DeptName | TotalEmployees | AvgSalary |
+-----+-----+-----+
| CSE      | 2              | 65000.0000 |
| IT       | 2              | 46500.0000 |
| ECE      | 1              | 52000.0000 |
+-----+-----+-----+
3 rows in set (0.02 sec)
```

7.

ANS : $\pi \text{ SID (Enroll)} \div \pi \text{ CID (Course)}$

8.

```
mysql> INSERT INTO Student VALUES (5, 'Neha', '2005-06-10', 101);
Query OK, 1 row affected (0.02 sec)

mysql> UPDATE Employee
-> SET Salary = Salary + 5000
-> WHERE DeptID = 101;
Query OK, 2 rows affected (0.02 sec)
Rows matched: 2 Changed: 2 Warnings: 0

mysql> DELETE FROM Student WHERE SID = 5;
Query OK, 1 row affected (0.02 sec)

mysql> select * from Student;
+----+-----+-----+-----+
| SID | Name  | DOB      | DeptID |
+----+-----+-----+-----+
| 1   | Riyaz | 2005-02-10 | 101    |
| 2   | Arjun | 2004-08-15 | 102    |
| 3   | Meena | 2005-01-20 | 101    |
| 4   | Sara  | 2004-12-05 | 103    |
+----+-----+-----+-----+
4 rows in set (0.00 sec)

mysql> select * from Employee;
+----+-----+-----+-----+
| EmpID | Name  | Salary | DeptID |
+----+-----+-----+-----+
| 1     | John  | 65000  | 101    |
| 2     | David | 45000  | 102    |
| 3     | Priya | 75000  | 101    |
| 4     | Ali   | 52000  | 103    |
| 5     | Kiran | 48000  | 102    |
+----+-----+-----+-----+
5 rows in set (0.00 sec)
```

9.

ANS : { e | Employee(e) $\wedge$ e.DeptID = 'IT' $\wedge$ e.Salary > 50000 }

10.

```
mysql> SELECT p.Name, f.Source, f.Destination
-> FROM Booking b
-> JOIN Passenger p ON b.PID = p.PID
-> JOIN Flight f ON b.FID = f.FID;
```

Name	Source	Destination
Ravi	Chennai	Delhi
Rahul	Chennai	Delhi
Anita	Mumbai	Bangalore
Ravi	Mumbai	Bangalore

4 rows in set (0.01 sec)

```
mysql>
mysql> -- 2. Subquery
mysql> SELECT Name FROM Passenger
-> WHERE PID IN (
-> SELECT PID FROM Booking WHERE FID = 101
-> );
```

Name
Ravi
Rahul

2 rows in set (0.01 sec)

```
mysql> CREATE VIEW FlightCount AS
-> SELECT FID, COUNT(*) AS TotalBookings
-> FROM Booking
-> GROUP BY FID;
Query OK, 0 rows affected (0.02 sec)
```

```
mysql>
mysql> SELECT * FROM FlightCount;
```

FID	TotalBookings
101	2
102	2

2 rows in set (0.00 sec)

```
mysql> -- 4. Another Join
mysql> SELECT p.Name, b.TravelDate
-> FROM Passenger p JOIN Booking b
-> ON p.PID = b.PID;
```

Name	TravelDate
Ravi	2026-08-01
Ravi	2026-08-03
Anita	2026-08-02
Rahul	2026-08-04

4 rows in set (0.00 sec)

```
mysql> -- 5. Aggregate
mysql> SELECT FID, COUNT(*) FROM Booking GROUP BY FID;
```

FID	COUNT(*)
101	2
102	2

2 rows in set (0.00 sec)